

DAY-01

DOCUMENTATION

Name: S.priya

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Team -04

Objective:

The objective of this task is to learn about Small Language Models (SLMs), their applications in real-world domains, and the importance of tokenizers. It also aims to understand how SLMs solve specific problems and how tokenization helps language models process and understand text efficiently.

- SLM:

1. What is SLM ?

A Small Language Model (SLM) is an AI language model with fewer parameters than Large Language Models (LLMs). SLMs are designed to perform language tasks efficiently while using less memory, computational power, and cost.

2. Key features of SLM ?

- Smaller in size compared to LLMs.
- Faster response and processing time.
- Requires less memory and computing resources.
- Can run on mobile devices, laptops, and edge devices.

3. Advantages of SLMs ?

- Reduced hardware requirements.
- Lower deployment and maintenance costs.
- Can work in environments with limited resources

4. Common Applications of SLMs ?

- Educational assistants.
- Customer support chatbots.
- Healthcare information systems.
- Smart mobile applications.

- **Selected Domain : Education.**

1. **Why I Chose Education**

Education is an important field where students often need quick access to learning support and doubt clarification. A Small Language Model (SLM) can provide personalized assistance while requiring fewer computational resources.

2. **Proposed SLM: Student Learning Assistant**

Problem It Solves:

- Helps students clear basic academic doubts.
- Provides quick explanations of concepts.
- Assists with homework and learning activities.

Why It Is Useful:

- Available anytime for students.
- Supports self-learning.
- Provides instant responses without waiting for a teacher.
- Can run on low-resource devices such as laptops and mobile phones.

Who Would Use It:

- College students.
- School students.
- Teachers.
- Online learners.

3. **What Problem It Solves:**

The Student Learning Assistant SLM helps students get quick answers to their academic questions. Many students face difficulties in understanding concepts, completing homework, and finding study materials when teachers are not available. This SLM provides instant explanations, doubt clarification, and learning

support, making education more accessible and effective.

4. Why It Is Useful in the Education Domain:

The Student Learning Assistant SLM is useful in the education domain because it provides students with quick and easy access to learning support. It helps students understand concepts, clear doubts instantly, and learn at their own pace. The model can be available anytime, making learning more accessible and efficient.

5. Who would use it:

- School Students- To learn concepts and clear doubts.
- College Students- To get academic support and study assistance.
- Teachers- To prepare notes and provide learning resources.
- Online Learners- To learn new topics and improve their knowledge.

6. What is a tokenizer?

A Tokenizer is a tool that breaks text into smaller units called tokens so that a language model can understand and process the text.

7. Why is tokenization needed?

Tokenization is needed because AI models cannot understand raw text directly. The text must first be broken into smaller tokens that the model can process.

Reasons for Tokenization:

- Converts text into a format that AI models can understand.
- Breaks large sentences into smaller manageable pieces.
- Helps the model analyze words and their meanings.
- Improves the efficiency of text processing.

8. Types of tokenizers:

There are mainly three common types of tokenizers:

1. Word Tokenizer- Splits text into words.
2. Character Tokenizer- Splits text into individual characters.

3. Subword Tokenizer- Splits words into smaller meaningful parts.

9. How tokenization impacts model performance:

Tokenization plays an important role in how efficiently a language model processes and understands text.

Impact on Model Performance:

- Improves Accuracy
- Increases Processing Speed
- Reduces Memory Usage
- Handles Unknown Words
- Affects Training Efficiency

Conclusion:

Small Language Models (SLMs) are useful AI models that can perform tasks quickly while using fewer resources. In the Education domain, they help students learn and clear doubts easily. Tokenization helps the model understand text better and improves its performance. Overall, SLMs are a simple, efficient, and cost-effective solution for many real-world applications.